

MA641 Research Project

Yash Tanaji Bhosale

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```
library(dplyr)
```

```
##  
## Attaching package: 'dplyr'  
  
## The following objects are masked from 'package:stats':  
##  
##   filter, lag  
  
## The following objects are masked from 'package:base':  
##  
##   intersect, setdiff, setequal, union
```

```
library(lubridate)
```

```
## Warning: package 'lubridate' was built under R version 4.5.2  
  
##  
## Attaching package: 'lubridate'  
  
## The following objects are masked from 'package:base':  
##  
##   date, intersect, setdiff, union
```

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.2
```

```
#load data  
gold_raw <- read.csv("XAU_USD Historical Data.csv", stringsAsFactors = FALSE)  
  
#cleaning and converting cols  
gold <- gold_raw %>%  
  mutate(  
    Date = mdy(Date), # convert to Date type  
    Price = as.numeric(gsub(",", "", Price)) # remove commas → convert to numeric  
  ) %>%  
  arrange(Date) %>%
```

```

select(Date, Price)

#summary of price col
summary(gold$Price)

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1052    1272    1690    1740    1930    4356

#Missing value check
sum(is.na(gold$Price))

## [1] 0

sum(is.na(gold$Date))

## [1] 0

#Plot the raw price series
ggplot(gold, aes(x = Date, y = Price)) +
  geom_line(color = "darkgoldenrod2") +
  labs(
    title = "Daily Gold Spot Price (XAU/USD), 2015-Present",
    x = "Date",
    y = "Gold Price (USD)"
  ) +
  theme_minimal()

```

Daily Gold Spot Price (XAU/USD), 2015–Present



```
library(moments)
```

```
## Warning: package 'moments' was built under R version 4.5.2
```

```
#Basic descriptive statistics
mean_price <- mean(gold$Price, na.rm = TRUE)
var_price  <- var(gold$Price, na.rm = TRUE)
sd_price   <- sd(gold$Price, na.rm = TRUE)
skew_price <- skewness(gold$Price, na.rm = TRUE)
kurt_price <- kurtosis(gold$Price, na.rm = TRUE)
```

```
cat("Mean:", mean_price, "\n")
```

```
## Mean: 1739.502
```

```
cat("Variance:", var_price, "\n")
```

```
## Variance: 370352.5
```

```
cat("Standard Deviation:", sd_price, "\n")
```

```
## Standard Deviation: 608.5659
```

```
cat("Skewness:", skew_price, "\n")
```

```
## Skewness: 1.583027
```

```
cat("Kurtosis:", kurt_price, "\n")
```

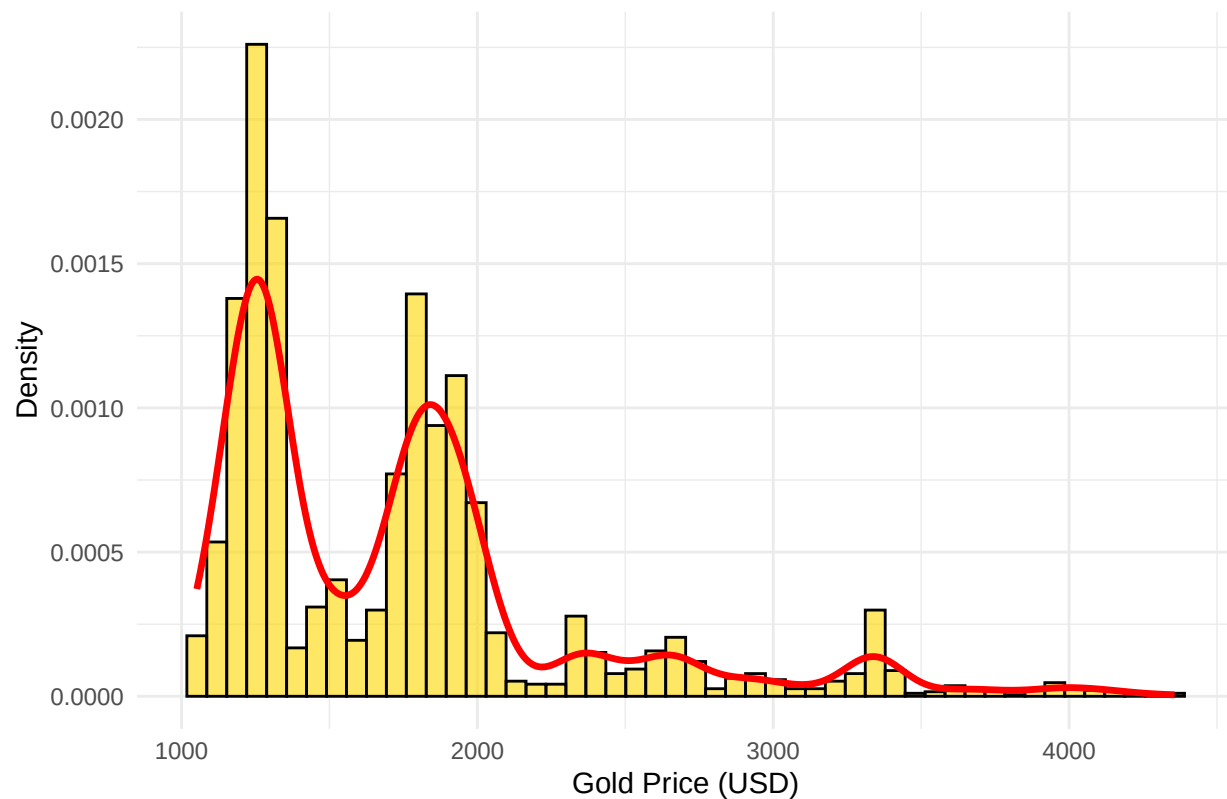
```
## Kurtosis: 5.570147
```

```
ggplot(gold, aes(x = Price)) +  
  geom_histogram(aes(y = ..density..),  
                 bins = 50,  
                 fill = "gold",  
                 alpha = 0.6,  
                 color = "black") +  
  geom_density(color = "red", size = 1.2) +  
  labs(title = "Histogram + Density of Gold Prices",  
       x = "Gold Price (USD)",  
       y = "Density") +  
  theme_minimal()
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.  
## i Please use 'linewidth' instead.  
## This warning is displayed once every 8 hours.  
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was  
## generated.
```

```
## Warning: The dot-dot notation ('..density..') was deprecated in ggplot2 3.4.0.  
## i Please use 'after_stat(density)' instead.  
## This warning is displayed once every 8 hours.  
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was  
## generated.
```

Histogram + Density of Gold Prices



```
library(tseries)
```

```
## Warning: package 'tseries' was built under R version 4.5.2
```

```
## Registered S3 method overwritten by 'quantmod':
##   method      from
##   as.zoo.data.frame zoo
```

```
library(zoo)
```

```
##
## Attaching package: 'zoo'
```

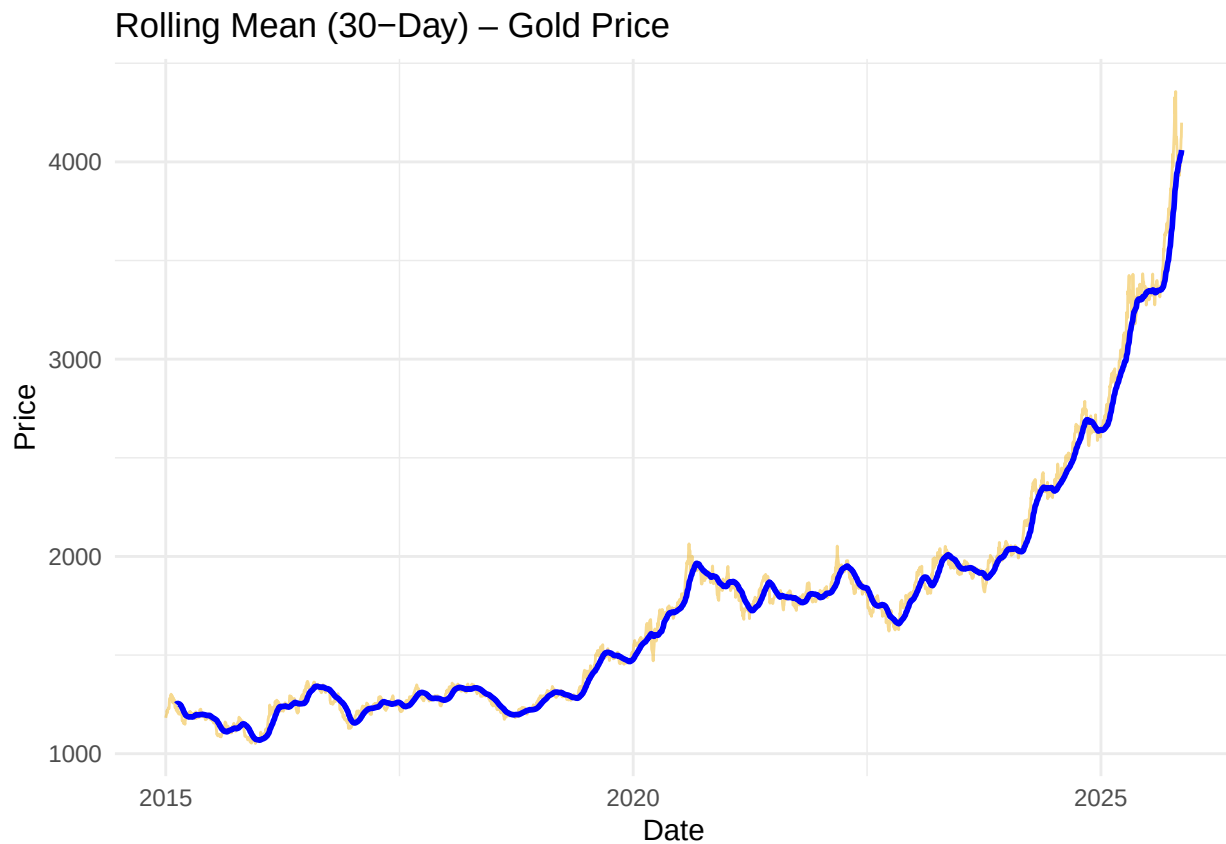
```
## The following objects are masked from 'package:base':
##
##   as.Date, as.Date.numeric
```

```
gold$roll_mean <- rollmean(gold$Price, k = 30, fill = NA, align = "right")
gold$roll_var  <- rollapply(gold$Price, width = 30, FUN = var, fill = NA, align = "right")

# 2.2 Plot rolling mean & variance
ggplot(gold, aes(x = Date)) +
  geom_line(aes(y = Price), color = "goldenrod2", alpha = 0.5) +
```

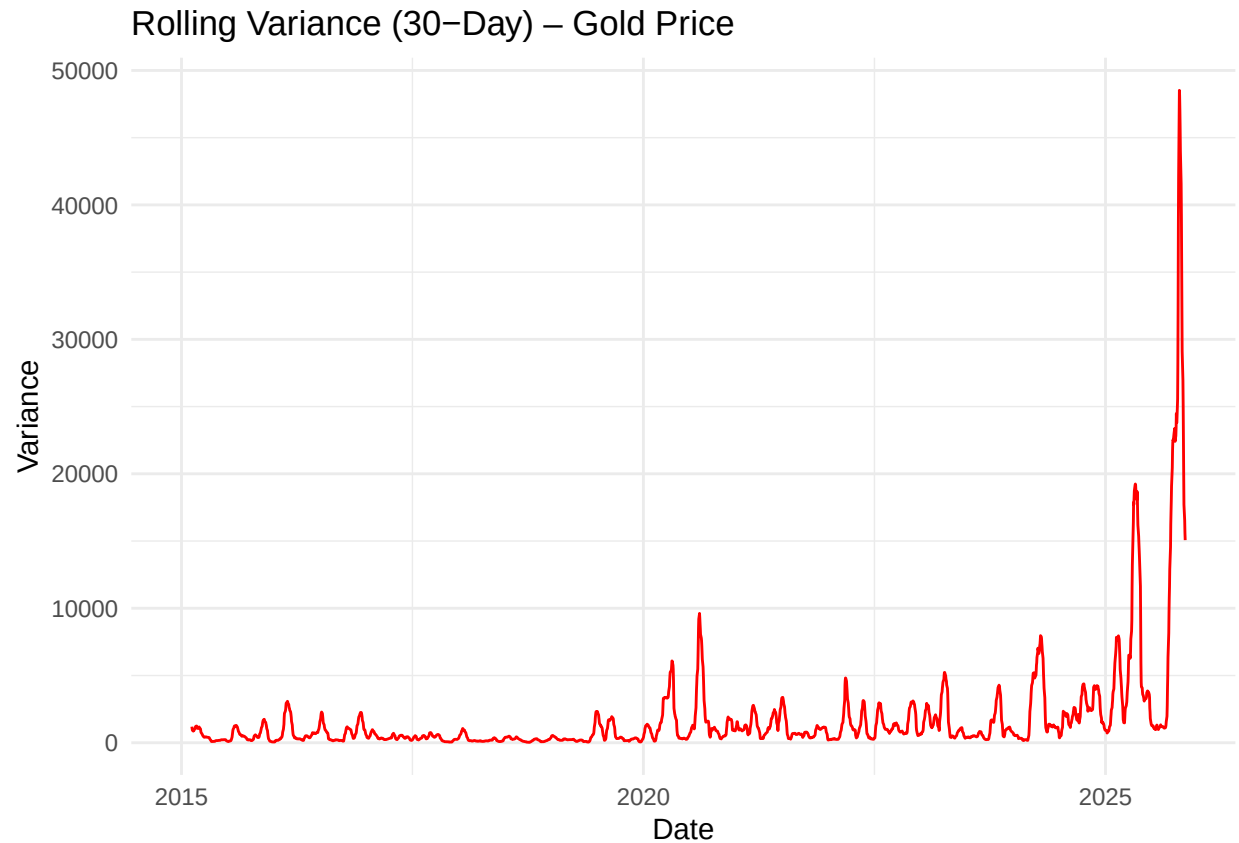
```
geom_line(aes(y = roll_mean), color = "blue", size = 1) +
labs(title = "Rolling Mean (30-Day) - Gold Price",
      y = "Price") +
theme_minimal()
```

```
## Warning: Removed 29 rows containing missing values or values outside the scale range
## ('geom_line()').
```



```
ggplot(gold, aes(x = Date, y = roll_var)) +
geom_line(color = "red") +
labs(title = "Rolling Variance (30-Day) - Gold Price",
      y = "Variance") +
theme_minimal()
```

```
## Warning: Removed 29 rows containing missing values or values outside the scale range
## ('geom_line()').
```



```
#ADF test on RAW PRICE
adf_test_raw <- adf.test(gold$Price, alternative = "stationary")
```

```
## Warning in adf.test(gold$Price, alternative = "stationary"): p-value greater
## than printed p-value
```

```
adf_test_raw
```

```
##
## Augmented Dickey-Fuller Test
##
## data: gold$Price
## Dickey-Fuller = 2.5797, Lag order = 14, p-value = 0.99
## alternative hypothesis: stationary
```

```
library(tseries)
library(ggplot2)
library(dplyr)
```

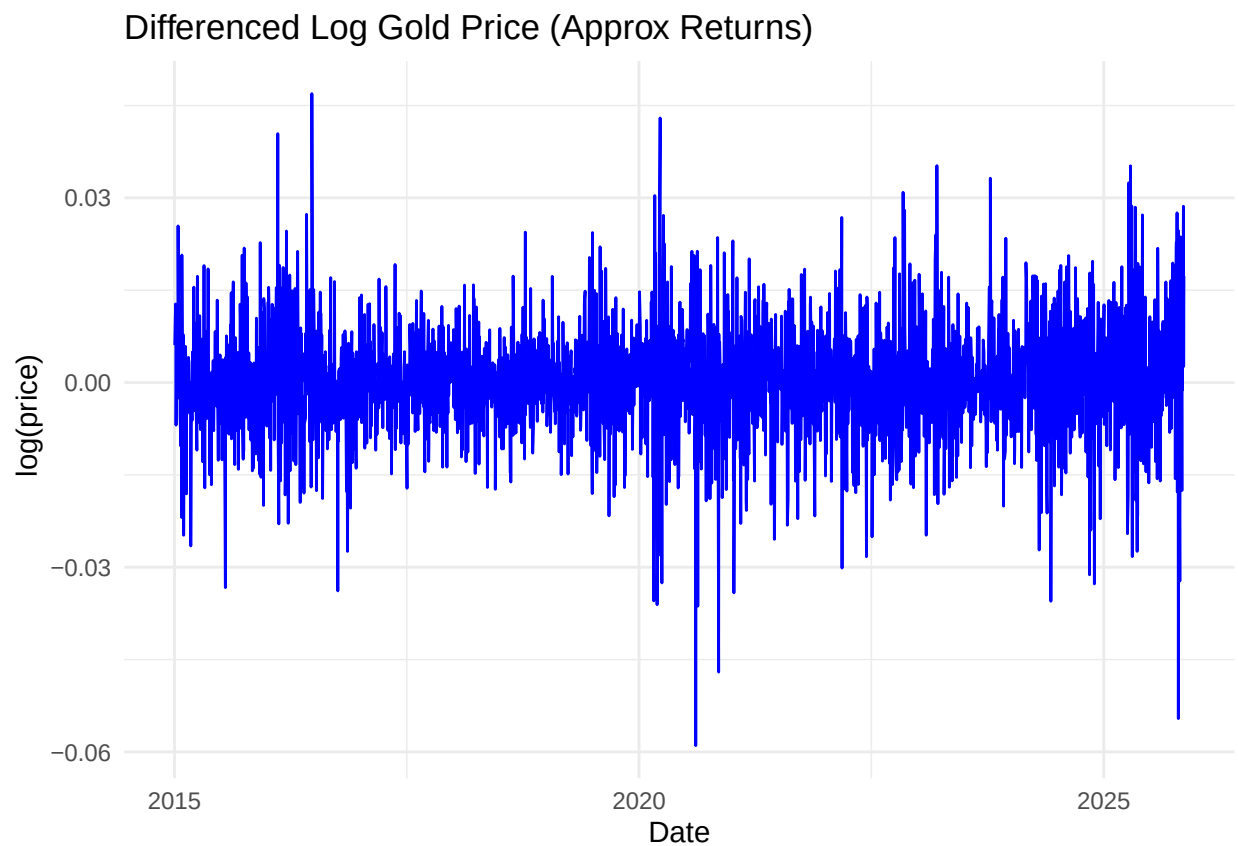
```
# 3.1 Apply log transform
gold$log_price <- log(gold$Price)
```

```
# 3.2 First differencing ( log(price))
gold <- gold %>%
```

```
mutate(
  diff_log_price = log_price - lag(log_price)
)

# 3.3 Create a clean dataset without the first NA row
gold_diff <- gold %>% filter(!is.na(diff_log_price))

# 3.4 Plot differenced series
ggplot(gold_diff, aes(x = Date, y = diff_log_price)) +
  geom_line(color = "blue") +
  labs(
    title = "Differenced Log Gold Price (Approx Returns)",
    x = "Date",
    y = "log(price)"
  ) +
  theme_minimal()
```



```
# 3.5 ADF test on differenced log prices
adf_test_diff <- adf.test(gold_diff$diff_log_price, alternative = "stationary")
```

```
## Warning in adf.test(gold_diff$diff_log_price, alternative = "stationary"):
## p-value smaller than printed p-value
```



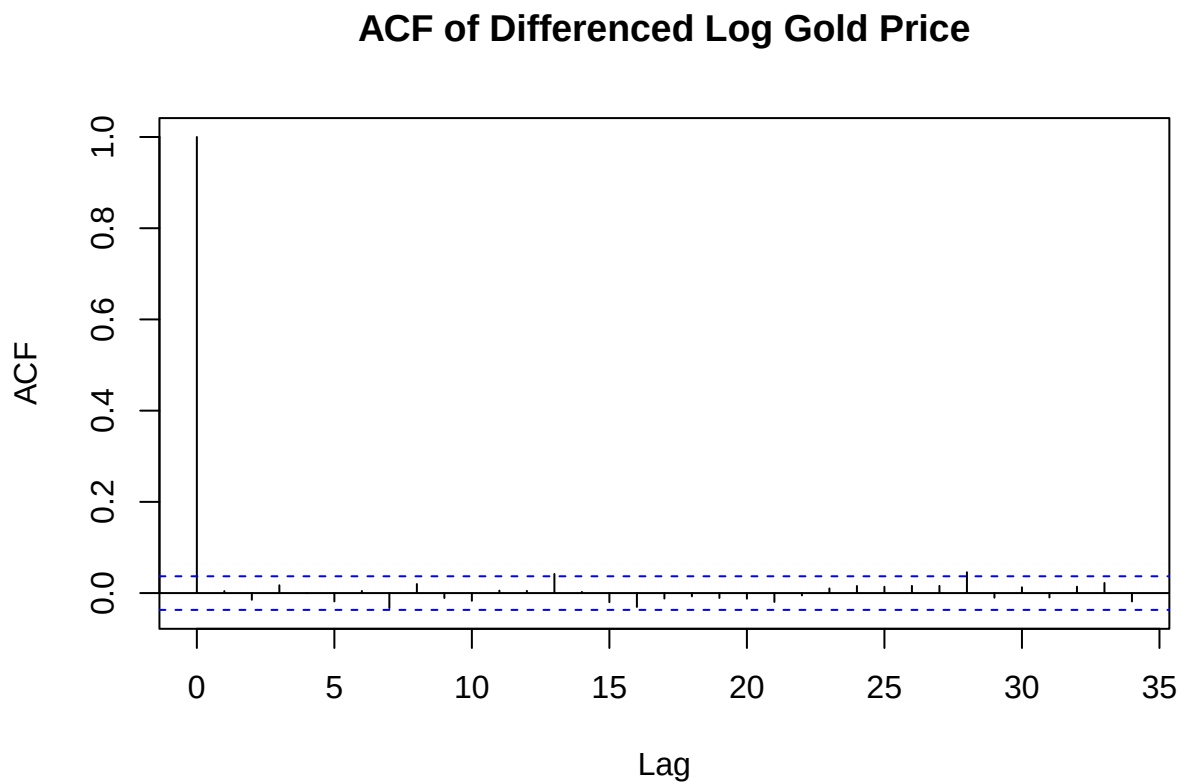
```
adf_test_diff
```

```
##  
## Augmented Dickey-Fuller Test  
##  
## data: gold_diff$diff_log_price  
## Dickey-Fuller = -13.991, Lag order = 14, p-value = 0.01  
## alternative hypothesis: stationary
```

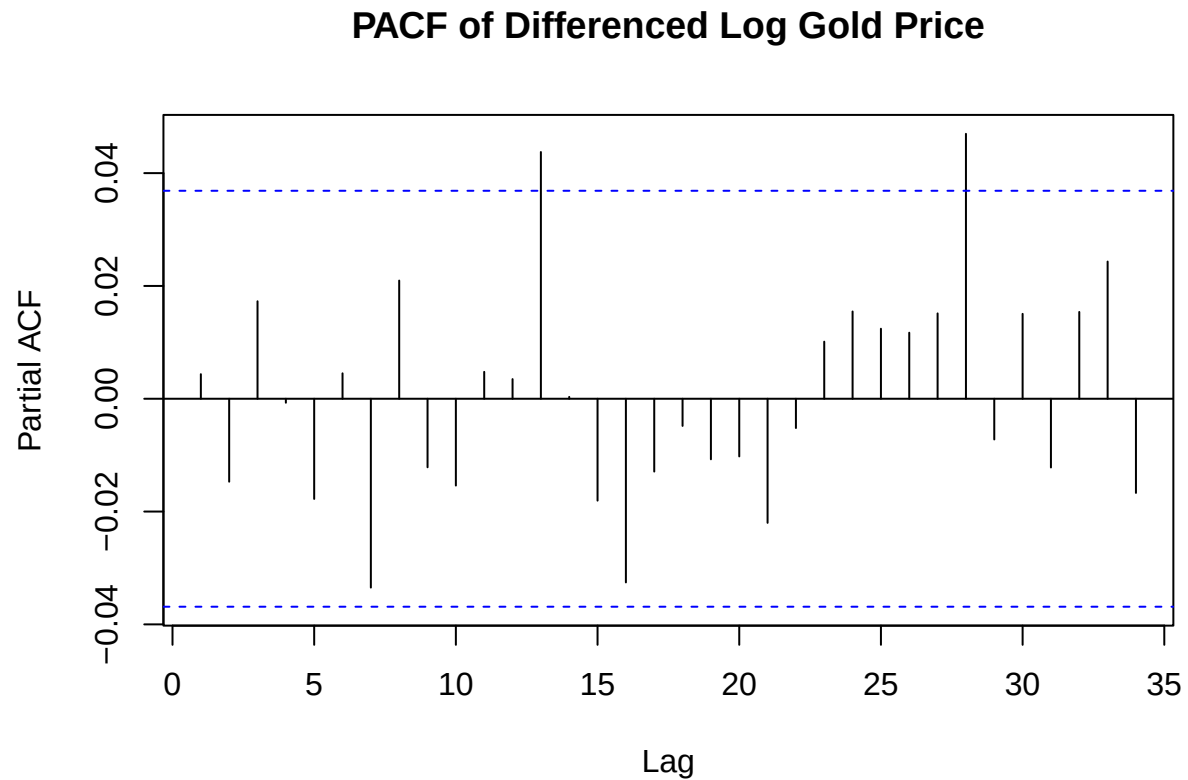
```
library(forecast)
```

```
##  
## Attaching package: 'forecast'  
  
## The following object is masked _by_ '.GlobalEnv':  
##  
## gold
```

```
# 4.1 ACF plot  
acf(gold_diff$diff_log_price,  
    main = "ACF of Differenced Log Gold Price")
```



```
# 4.2 PACF plot
pacf(gold_diff$diff_log_price,
     main = "PACF of Differenced Log Gold Price")
```



```
library(forecast)

# Use the stationary series
y <- gold_diff$diff_log_price

# 5.1 ARIMA(0,1,1) -- MA(1)
fit_011 <- Arima(gold$Price, order = c(0,1,1))
summary(fit_011)

## Series: gold$Price
## ARIMA(0,1,1)
##
## Coefficients:
##          ma1
##        -0.0101
## s.e.    0.0186
##
## sigma^2 = 353.3: log likelihood = -12299.94
## AIC=24603.87  AICc=24603.88  BIC=24615.77
##
## Training set error measures:
```

```
##               ME      RMSE      MAE      MPE      MAPE      MASE
## Training set 1.078173 18.78962 12.06872 0.04121075 0.658304 0.9992522
##               ACF1
## Training set -0.003376058
```

```
# 5.2 ARIMA(1,1,0) -- AR(1)
fit_110 <- Arima(gold$Price, order = c(1,1,0))
summary(fit_110)
```

```
## Series: gold$Price
## ARIMA(1,1,0)
##
## Coefficients:
##      ar1
##      -0.0104
## s.e.    0.0189
##
## sigma^2 = 353.3: log likelihood = -12299.93
## AIC=24603.87 AICc=24603.87 BIC=24615.76
##
## Training set error measures:
##               ME      RMSE      MAE      MPE      MAPE      MASE
## Training set 1.078392 18.78959 12.06859 0.04121914 0.6582988 0.9992418
##               ACF1
## Training set -0.003060185
```

```
# 5.3 ARIMA(1,1,1) -- ARMA mixed
fit_111 <- Arima(gold$Price, order = c(1,1,1))
summary(fit_111)
```

```
## Series: gold$Price
## ARIMA(1,1,1)
##
## Coefficients:

## Warning in sqrt(diag(x$var.coef)): NaNs produced

##      ar1      ma1
##      -0.0052 -0.005
## s.e.      NaN      NaN
##
## sigma^2 = 353.4: log likelihood = -12299.93
## AIC=24605.87 AICc=24605.88 BIC=24623.71
##
## Training set error measures:
##               ME      RMSE      MAE      MPE      MAPE      MASE
## Training set 1.078219 18.78961 12.06868 0.04121249 0.6583027 0.9992492
##               ACF1
## Training set -0.003279205
```

```

# 5.4 ARIMA(0,1,0) -- Random Walk baseline
fit_010 <- Arima(gold$Price, order = c(0,1,0))
summary(fit_010)

## Series: gold$Price
## ARIMA(0,1,0)
##
## sigma^2 = 353.2: log likelihood = -12300.09
## AIC=24602.17 AICc=24602.17 BIC=24608.12
##
## Training set error measures:
##           ME      RMSE      MAE      MPE      MAPE      MASE
## Training set 1.067542 18.79061 12.0739 0.04080138 0.6585639 0.9996809
##           ACF1
## Training set -0.01355035

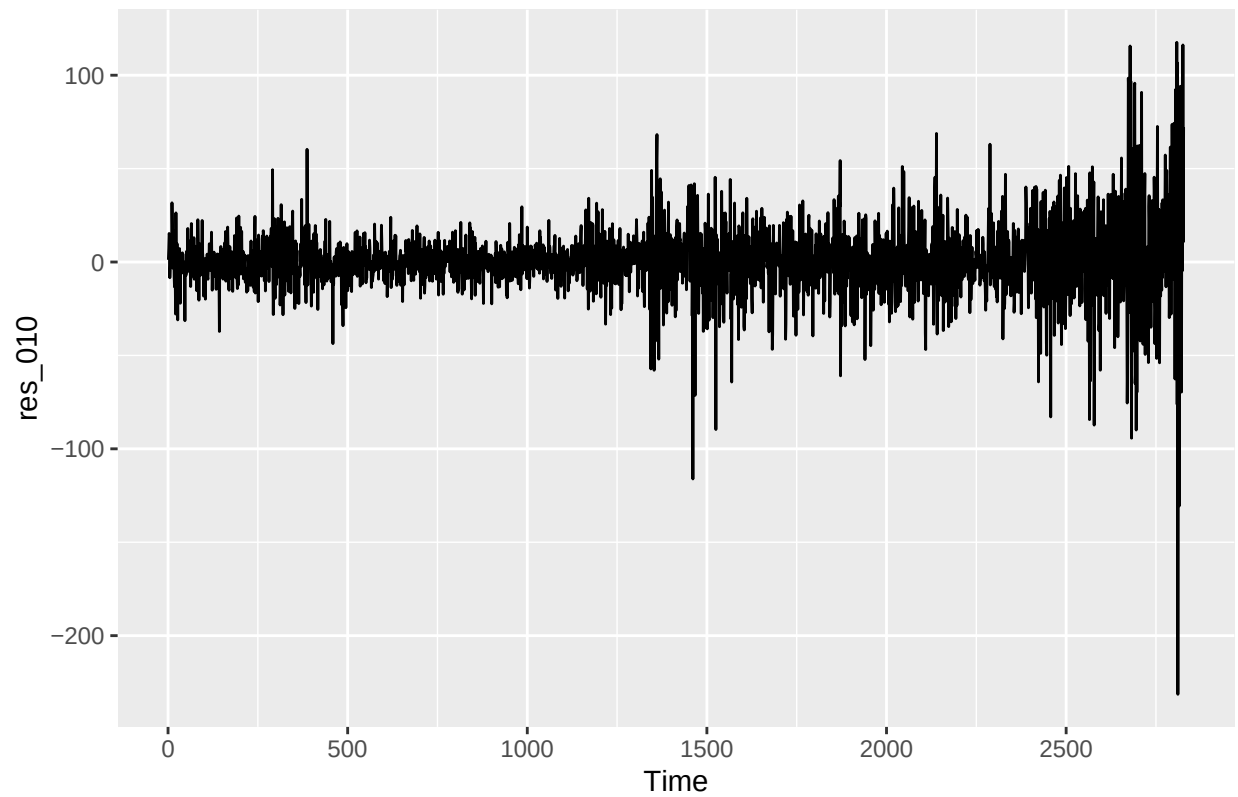
library(forecast)

# Extract residuals
res_010 <- residuals(fit_010)
res_011 <- residuals(fit_011)
res_110 <- residuals(fit_110)

# 6.1 Plot residuals (random noise?)
autoplot(res_010) +
  ggtitle("Residuals of ARIMA(0,1,0)")

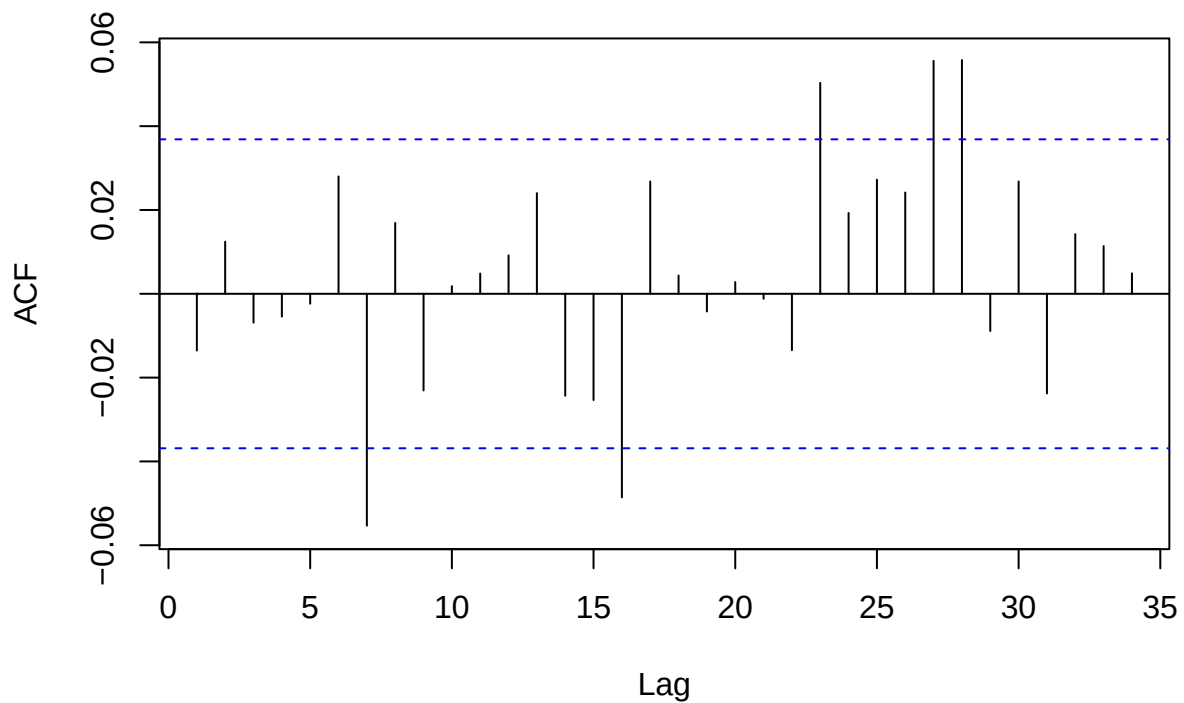
```

Residuals of ARIMA(0,1,0)



```
# 6.2 ACF of residuals (should have no significant spikes)  
Acf(res_010, main = "ACF of Residuals - ARIMA(0,1,0)")
```

ACF of Residuals — ARIMA(0,1,0)



```
# 6.3 Ljung-Box test
```

```
Box.test(res_010, lag = 20, type = "Ljung-Box")
```

```
##
```

```
## Box-Ljung test
```

```
##
```

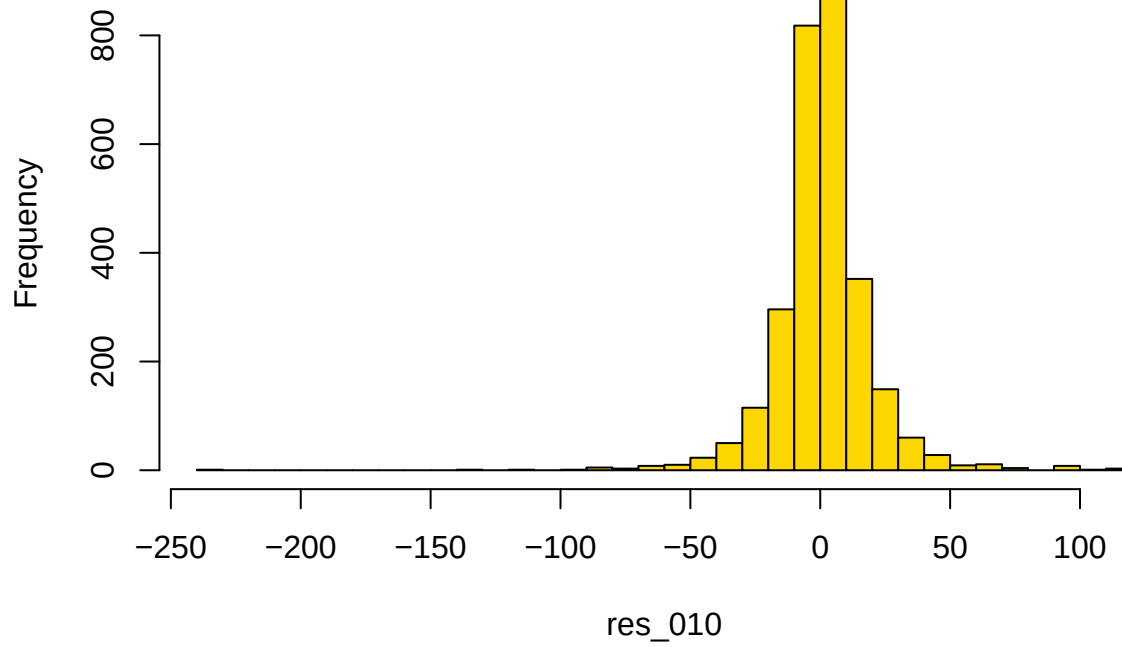
```
## data: res_010
```

```
## X-squared = 28.775, df = 20, p-value = 0.09228
```

```
#6.4 Histogram of residuals
```

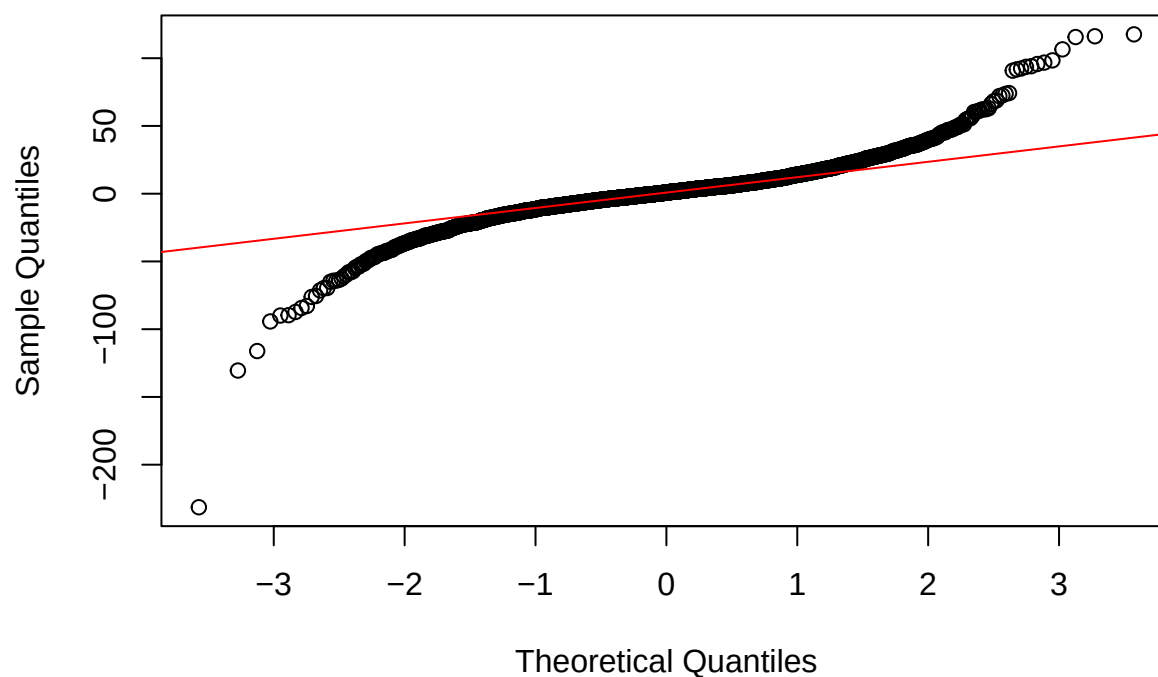
```
hist(res_010, breaks = 40, col = "gold", main = "Histogram of Residuals")
```

Histogram of Residuals



```
# 6.5 Q-Q plot (should be roughly straight line)
qqnorm(res_010)
qqline(res_010, col = "red")
```

Normal Q-Q Plot



```
library(forecast)

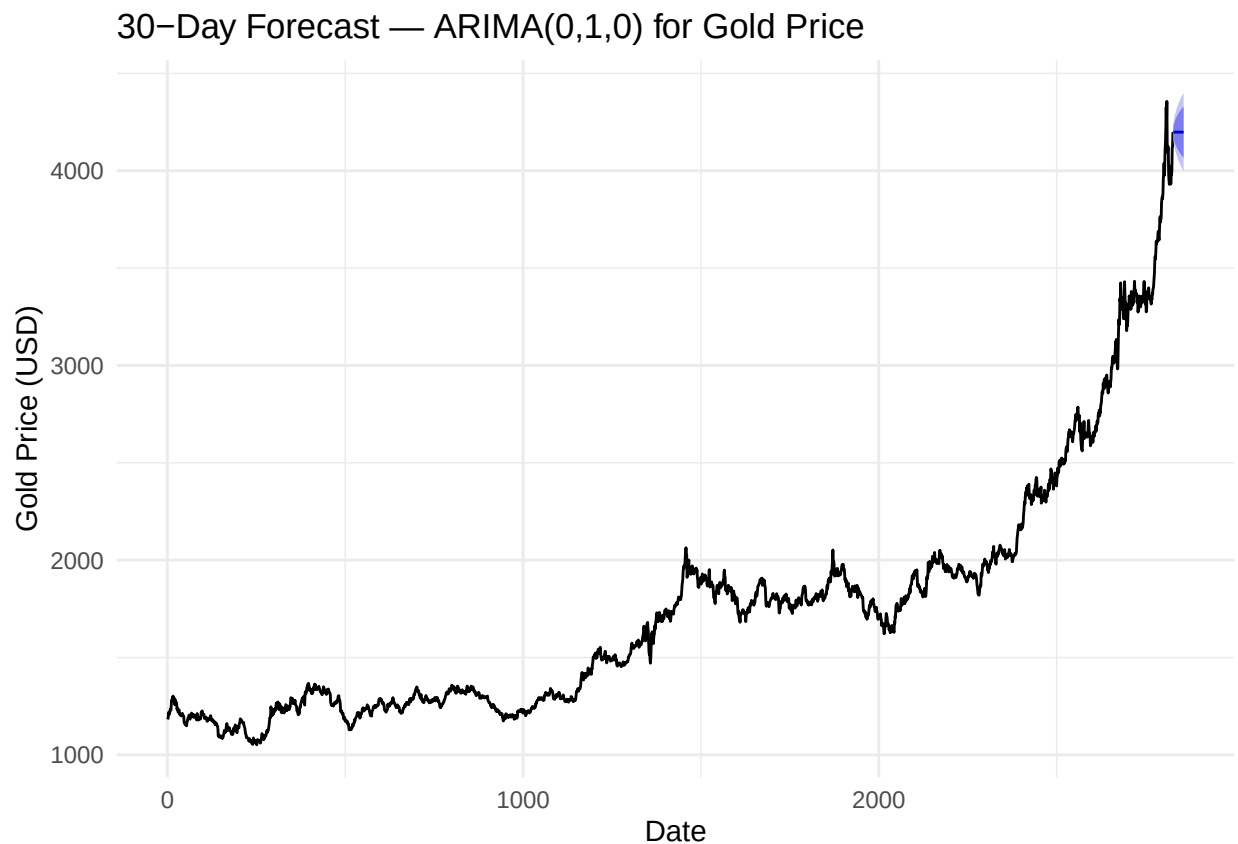
# Use the final model fit_010 from Step 5
# Forecast next 30 days
forecast_30 <- forecast(fit_010, h = 30)

# Print forecast values
forecast_30
```

##	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
## 2828	4198.73	4174.645	4222.815	4161.895	4235.565
## 2829	4198.73	4164.668	4232.792	4146.637	4250.823
## 2830	4198.73	4157.013	4240.447	4134.929	4262.531
## 2831	4198.73	4150.559	4246.901	4125.059	4272.401
## 2832	4198.73	4144.873	4252.587	4116.363	4281.097
## 2833	4198.73	4139.733	4257.727	4108.502	4288.958
## 2834	4198.73	4135.006	4262.454	4101.273	4296.187
## 2835	4198.73	4130.606	4266.854	4094.544	4302.916
## 2836	4198.73	4126.474	4270.986	4088.224	4309.236
## 2837	4198.73	4122.565	4274.895	4082.246	4315.214
## 2838	4198.73	4118.848	4278.612	4076.561	4320.899
## 2839	4198.73	4115.296	4282.164	4071.128	4326.332
## 2840	4198.73	4111.889	4285.571	4065.918	4331.542
## 2841	4198.73	4108.611	4288.849	4060.904	4336.556
## 2842	4198.73	4105.448	4292.012	4056.067	4341.393


```
## 2843      4198.73 4102.388 4295.072 4051.388 4346.072
## 2844      4198.73 4099.423 4298.037 4046.854 4350.606
## 2845      4198.73 4096.544 4300.916 4042.451 4355.009
## 2846      4198.73 4093.744 4303.716 4038.168 4359.292
## 2847      4198.73 4091.017 4306.443 4033.997 4363.463
## 2848      4198.73 4088.357 4309.103 4029.929 4367.531
## 2849      4198.73 4085.760 4311.700 4025.957 4371.503
## 2850      4198.73 4083.221 4314.239 4022.074 4375.386
## 2851      4198.73 4080.736 4316.724 4018.274 4379.186
## 2852      4198.73 4078.303 4319.157 4014.553 4382.907
## 2853      4198.73 4075.918 4321.542 4010.905 4386.555
## 2854      4198.73 4073.579 4323.881 4007.328 4390.132
## 2855      4198.73 4071.282 4326.178 4003.815 4393.645
## 2856      4198.73 4069.026 4328.434 4000.365 4397.095
## 2857      4198.73 4066.809 4330.651 3996.974 4400.486
```

```
# Plot forecast
autoplot(forecast_30) +
  labs(
    title = "30-Day Forecast - ARIMA(0,1,0) for Gold Price",
    x = "Date",
    y = "Gold Price (USD)"
  ) +
  theme_minimal()
```



```
library(forecast)

# Use the final model fit_010 from Step 5
# Forecast next 60 days
forecast_60 <- forecast(fit_010, h = 60)

# Print forecast values
forecast_60
```

##	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
## 2828	4198.73	4174.645	4222.815	4161.895	4235.565
## 2829	4198.73	4164.668	4232.792	4146.637	4250.823
## 2830	4198.73	4157.013	4240.447	4134.929	4262.531
## 2831	4198.73	4150.559	4246.901	4125.059	4272.401
## 2832	4198.73	4144.873	4252.587	4116.363	4281.097
## 2833	4198.73	4139.733	4257.727	4108.502	4288.958
## 2834	4198.73	4135.006	4262.454	4101.273	4296.187
## 2835	4198.73	4130.606	4266.854	4094.544	4302.916
## 2836	4198.73	4126.474	4270.986	4088.224	4309.236
## 2837	4198.73	4122.565	4274.895	4082.246	4315.214
## 2838	4198.73	4118.848	4278.612	4076.561	4320.899
## 2839	4198.73	4115.296	4282.164	4071.128	4326.332
## 2840	4198.73	4111.889	4285.571	4065.918	4331.542
## 2841	4198.73	4108.611	4288.849	4060.904	4336.556
## 2842	4198.73	4105.448	4292.012	4056.067	4341.393
## 2843	4198.73	4102.388	4295.072	4051.388	4346.072
## 2844	4198.73	4099.423	4298.037	4046.854	4350.606
## 2845	4198.73	4096.544	4300.916	4042.451	4355.009
## 2846	4198.73	4093.744	4303.716	4038.168	4359.292
## 2847	4198.73	4091.017	4306.443	4033.997	4363.463
## 2848	4198.73	4088.357	4309.103	4029.929	4367.531
## 2849	4198.73	4085.760	4311.700	4025.957	4371.503
## 2850	4198.73	4083.221	4314.239	4022.074	4375.386
## 2851	4198.73	4080.736	4316.724	4018.274	4379.186
## 2852	4198.73	4078.303	4319.157	4014.553	4382.907
## 2853	4198.73	4075.918	4321.542	4010.905	4386.555
## 2854	4198.73	4073.579	4323.881	4007.328	4390.132
## 2855	4198.73	4071.282	4326.178	4003.815	4393.645
## 2856	4198.73	4069.026	4328.434	4000.365	4397.095
## 2857	4198.73	4066.809	4330.651	3996.974	4400.486
## 2858	4198.73	4064.628	4332.832	3993.639	4403.821
## 2859	4198.73	4062.482	4334.978	3990.357	4407.103
## 2860	4198.73	4060.370	4337.090	3987.127	4410.333
## 2861	4198.73	4058.289	4339.171	3983.944	4413.516
## 2862	4198.73	4056.239	4341.221	3980.809	4416.651
## 2863	4198.73	4054.218	4343.242	3977.717	4419.743
## 2864	4198.73	4052.224	4345.236	3974.669	4422.791
## 2865	4198.73	4050.258	4347.202	3971.661	4425.799
## 2866	4198.73	4048.317	4349.143	3968.693	4428.767
## 2867	4198.73	4046.401	4351.059	3965.762	4431.698
## 2868	4198.73	4044.508	4352.952	3962.868	4434.592
## 2869	4198.73	4042.639	4354.821	3960.009	4437.451
## 2870	4198.73	4040.792	4356.668	3957.184	4440.276

```
## 2871      4198.73 4038.966 4358.494 3954.391 4443.069
## 2872      4198.73 4037.160 4360.300 3951.630 4445.830
## 2873      4198.73 4035.375 4362.085 3948.900 4448.560
## 2874      4198.73 4033.609 4363.851 3946.199 4451.261
## 2875      4198.73 4031.862 4365.598 3943.527 4453.933
## 2876      4198.73 4030.132 4367.328 3940.882 4456.578
## 2877      4198.73 4028.421 4369.039 3938.264 4459.196
## 2878      4198.73 4026.726 4370.734 3935.672 4461.788
## 2879      4198.73 4025.048 4372.412 3933.106 4464.354
## 2880      4198.73 4023.386 4374.074 3930.564 4466.896
## 2881      4198.73 4021.739 4375.721 3928.046 4469.414
## 2882      4198.73 4020.108 4377.352 3925.551 4471.909
## 2883      4198.73 4018.491 4378.969 3923.079 4474.381
## 2884      4198.73 4016.889 4380.571 3920.629 4476.831
## 2885      4198.73 4015.301 4382.159 3918.200 4479.260
## 2886      4198.73 4013.727 4383.733 3915.792 4481.668
## 2887      4198.73 4012.165 4385.295 3913.404 4484.056
```

```
# Plot forecast
autoplot(forecast_60) +
  labs(
    title = "60-Day Forecast - ARIMA(0,1,0) for Gold Price",
    x = "Date",
    y = "Gold Price (USD)"
  ) +
  theme_minimal()
```

